Lab 3 Report

Proportional Control

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1 Introduction

1.1 Background and Context

In control systems engineering, compensators are essential tools used to enhance both the transient and steady-state performance of dynamic systems. In previous experiments, the kitticopter system was modeled and controlled using a simple proportional controller. Although this configuration provided basic stability, it exhibited significant limitations such as a large steady-state tracking error, sluggish response, and a relatively long settling time. To overcome these shortcomings, a more sophisticated control strategy is required. This experiment therefore focuses on the design and implementation of a lead compensator, which is known to improve the speed of response, increase phase margin, and reduce overshoot. By introducing a phase lead at appropriate frequencies, the compensator shifts the root locus toward the left half-plane, resulting in faster and better-damped system dynamics. The experiment extends the previous modeling work by applying a compensator design methodology based on the root locus approach. Both simulation and hardware implementation are performed to evaluate the compensator's effect on system stability and performance.

1.2 Objective of the Experiment

The primary objective of this laboratory exercise is to design, implement, and validate a lead compensator for the kitticopter system that meets specific time-domain performance targets. The compensator aims to:

- Reduce steady-state tracking error and improve altitude setpoint accuracy to within 0.4 m of the desired final value,
- Limit the peak overshoot to below 20%,
- Achieve a settling time (t_s) within 2% of the final value and less than the specified target of 8 s.

Through simulation and experimental validation, the designed compensator will be evaluated to ensure it achieves the desired transient response while maintaining adequate stability margins.

1.3 Scope of the Experiment

The scope of this work encompasses the following:

- Designing a lead compensator using the root locus method based on the identified system model parameters ($A = 13.736$, $\tau = 9.7589$ s, and $K_s = 0.98$ V/m),
- Simulating the compensated and uncompensated system responses in MATLAB to assess improvements in overshoot, settling time, and steady-state error,
- Physically implementing the designed compensator using an operational amplifier (op-amp) with resistor–capacitor (RC) networks,

- Experimentally validating the performance of the implemented compensator on the actual kitticopter system, and comparing the results against both theoretical predictions and the earlier proportional control performance.

This experiment not only demonstrates the advantages of using lead compensation to improve the kitticopter's dynamic performance but also highlights practical aspects such as hardware non-idealities, component tolerances, and real-world control limitations.

2 System Modeling and Proportional Control Performance

In the previous lab, the kitticopter system was represented as a second-order dynamic system, with its vertical motion influenced by rotor thrust, gravitational force, and aerodynamic drag. Using Newton's Second Law and assuming vertical motion only, the system dynamics were described by the differential equation:

$$\ddot{y}(t) = f_t(t) - b\dot{y}(t) - g, \tag{1}$$

where $y(t)$ is the vertical position, $f_t(t)$ is the thrust input, b is the aerodynamic damping coefficient, and g is the gravitational acceleration. Taking the Laplace Transform and assuming zero initial conditions, the open-loop transfer function from thrust input to altitude was derived as:

$$G_p(s) = \frac{A}{s(\tau s + 1)}, \tag{2}$$

with system parameters identified from experimental data as $A = 13.736$, $\tau = 9.7589$ s, and sensor gain $K_s = 0.98\ V/m$.

2.1 Proportional Control Performance

A proportional controller was implemented to regulate altitude by scaling the thrust input with a gain K_p. While this controller provided some level of setpoint tracking, its performance exhibited notable limitations.

Key characteristics observed in the step response for a voltage increase from 2.5 V to 5 V included:

- **Overshoot:** The system exceeded the 5 V setpoint, peaking at approximately 6.5 V. which is above the the allowable 30% overshoot, so further reduction would improve performance.
- **Settling Time:** The system required over 70 s to settle, significantly exceeding the desired response time, highlighting sluggish transient behavior.

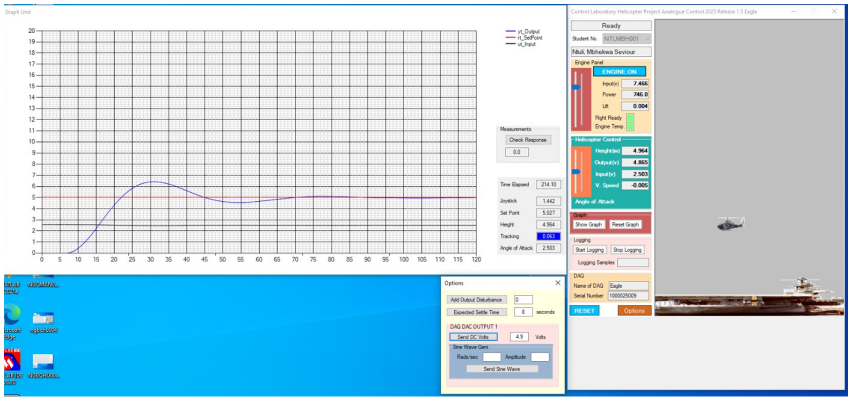

Figure 1: Step response of the kitticopter system under proportional control.

- **Tracking Accuracy:** The final output reached 4.964 V, corresponding to a tracking error of 6.3%, which is below the target maximum error of 8%.

While the proportional controller could approximately follow the setpoint, it did not meet design expectations for transient response and steady-state accuracy:

- The tracking error exceeded the 8% limit, indicating insufficient steady-state performance.
- The prolonged settling time revealed that the system responds slowly to input changes.
- Overshoot, though within the limit, could be further minimized for improved control.

These observations demonstrate the need for a more advanced control strategy. Consequently, a lead compensator is proposed in the following sections to enhance both transient response and tracking accuracy while maintaining system stability.

3 Controller Design

3.1 Choice of Controller

A lead compensator was selected for the kitticopter system because it improves both transient and steady-state performance while maintaining implementation simplicity. The proportional controller designed in Lab 2 resulted in a slow transient response and steady state tracking error of approximately 6.3%. The lead network adds a phase lead in the mid-frequency range, which effectively shifts the root locus to the left of the s-plane, yielding a faster and more stable response.

The main benefits motivating the choice of a lead compensator are:

- Enhanced transient response: Adds phase lead that reduces overshoot and settling time.
- Improved tracking accuracy: Increases system bandwidth, reducing steady state error.
- Simplicity and robustness: Offers an effective balance between performance and ease of physical implementation.
- Improved phase margin: Increases robustness to component tolerances and external disturbances.

The general compensator structure is

$$C(s) = K\frac{s+z}{s+p}, \quad z < p, \tag{3}$$

where K is the gain, z is the compensator zero, and p is its pole. Proper selection of z and p determines the phase lead and system damping improvement.

3.2 Design Specifications

The compensator was designed to meet the following time-domain requirements:

- Tracking accuracy: Altitude error ≤ 0.4 m ($\leq 8\%$ steady-state error).
- Settling time: $t_s \leq 8$ s (to within 2% of the final value).

The kitticopter's open-loop transfer function from Lab 2 is

$$G(s) = \frac{13.736}{s(9.7589s+1)}, \quad K_s = 0.98. \tag{4}$$

Since the system includes an integrator, its steady-state error to a step input ideally tends to zero for any finite gain. The design therefore focuses on improving transient response while maintaining this desirable steady-state property.

3.2.1 Tracking Accuracy $\geq$ 92%

The system must track the setpoint within 0.4 m of the final value, corresponding to a maximum tracking error of 8%. Using the sensor gain $K_s = 0.98$, the open-loop plant transfer function is:

$$G(s) = \frac{13.736}{s(9.7589s + 1)}. \tag{5}$$

The steady-state error for a step input is computed using the Final Value Theorem:

$$e_{ss} = \lim_{s \to 0} \frac{s}{1 + C(0)G(0)K_s}. \tag{6}$$

Since the plant contains an integrator:

$$G(0) = \frac{13.736}{0 \cdot (9.7589 \cdot 0 + 1)} = \infty \tag{7}$$

Thus:

$$e_{ss} = \frac{1}{1 + C(0) \cdot \infty \cdot K_s} = 0 \tag{8}$$

This guarantees 100% tracking accuracy for any positive compensator gain $K \geq 0$, satisfying the requirement.

3.2.2 Overshoot < 20%

To limit the overshoot, the damping ratio ζ is calculated as:

$$\zeta = -\frac{\ln(0.2)}{\sqrt{\pi^2 + (\ln(0.2))^2}} \approx 0.456 \tag{9}$$

The corresponding phase angle:

$$\theta = \cos^{-1}(\zeta) \approx 62.87^\circ \tag{10}$$

3.2.3 Settling Time $\leq$ 8 s

The expected settling time from simulation is $t_e = 8$ s. For a 2% criterion:

$$\sigma \leq 9/4 \approx 0.444 \tag{11}$$

The imaginary component of the critical pole:

$$\omega_y = \sigma \tan(\theta) = 0.444 \cdot \tan(62.87^\circ) \approx 0.867 \tag{12}$$

The critical pole location on the root locus is therefore:

$$s_c = -0.444 + j0.867 \tag{13}$$

3.2.4 Angle Criterion at the Critical Pole

In the root locus method, the angle criterion ensures that the sum of the angles contributed by all poles and zeros to a point on the root locus satisfies:

$$\sum \theta_{poles} - \sum \theta_{zeros} = (2k+1)180^\circ \tag{14}$$

Angles from poles: $\vartheta_2 = 180^\circ - \arctan \frac{0.867}{-0.444-(-0.925)} = 112.09^\circ$
$\theta_1 = 180^\circ - \arctan \frac{0.867}{-0.444} = 117.14^\circ$

Since these are poles, they contribute negative angles:

$$\theta_{total} = -117.14^\circ - 112.09^\circ = -229.23^\circ \tag{15}$$

The nearest multiple of 180° to -229.23° is -180°, requiring an additional phase lead:

$$\phi_{lead} = -180^\circ - (-229.23^\circ) = 49.23^\circ \tag{16}$$

3.2.5 Zero and Pole Locations

Using the calculated phase lead $\phi_{lead} = 49.23^\circ$, the zero location is:

$$z = \frac{-0.444 - 0.867}{\tan(49.23^\circ)} \approx -1.05 \tag{17}$$

The angle contributed by the zero is:

$$\theta_z = \arctan \frac{0.867}{1.05 - 0.444} = 55.07^\circ \tag{18}$$

The required angle from the pole to satisfy the overall phase criterion:

$$\theta_p = \theta_z - \phi_{lead} = 5.838^\circ \tag{19}$$

The pole location is then:

$$p = \frac{-0.444 - 0.867}{\tan(5.838^\circ)} \approx -8.92 \tag{20}$$

Thus, the lead compensator transfer function becomes:

$$C(s) = \frac{K(s+1.05)}{s+8.92} \tag{21}$$

3.2.6 Finding the Gain Value using MATLAB's SISO Tool

MATLAB's SISO Tool was used to determine the appropriate compensator gain. The plant transfer function was entered as:

```
G = tf([13.736], [9.7589 1 0]);
```

The root locus was plotted, and the gain was fine-tuned to:

$$K = 15 \tag{22}$$

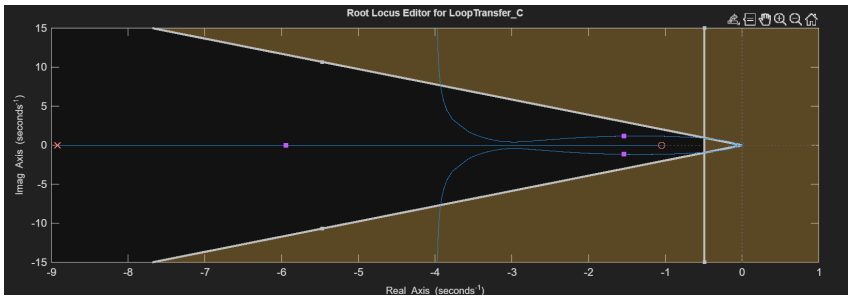

Figure 2: Root locus of the system with lead compensator zero and pole placement.

```
Tunable Block
Name: C
Sample Time: 0
Value:
   15.002 (s+1.05)
   ---------------
       (s+8.92)
```

Figure 3: Lead compensator transfer function with gain $K = 15$.

3.3 Validating the Compensator Performance

The system was simulated in MATLAB/Simulink using a step input of 2.5 V to track a 5 V output.

3.4 Lead Compensator Implementation

The lead compensator was constructed using a three-stage op-amp configuration, including:

1. An inverting amplifier to scale the input.
2. The compensator network containing resistors and capacitors to introduce the required phase lead.
3. A unity-gain buffer to stabilize the output.

Component selection: $\mathrm{R}_1 = R_2 = 2.7k\Omega, C_1 = C_2 = 100\mu F$
$R_4 \approx 10k\Omega, R_5 \approx 1.1k\Omega$
$R_3 \approx 73k\Omega, R_6 = R_7 = R_8 = 100k\Omega$

4 Controller Evaluation

4.1 Tracking Specification

The response of the system tracks the input at approximately 4.9 V, resulting in a tracking error of:

$$TrackingError(\%) = \frac{5 - 4.9}{5} \times 100 = 2\%$$

This tracking error of 2% = 0.1 m, satisfies the specification of tracking within 0.4 m of the final value, as the error is below the acceptable threshold.

4.2 Overshoot Specification

The overshoot specification requires the overshoot to be less than 20%. Based on the blue response curve, the peak value reached is approximately 5.2 V. The overshoot is calculated as:

$$Overshoot(\%) = \frac{5.2 - 5}{5} \times 100 = 4\%$$

This 4% overshoot is well within the specification, meeting the requirement for overshoot to be less than 20%.

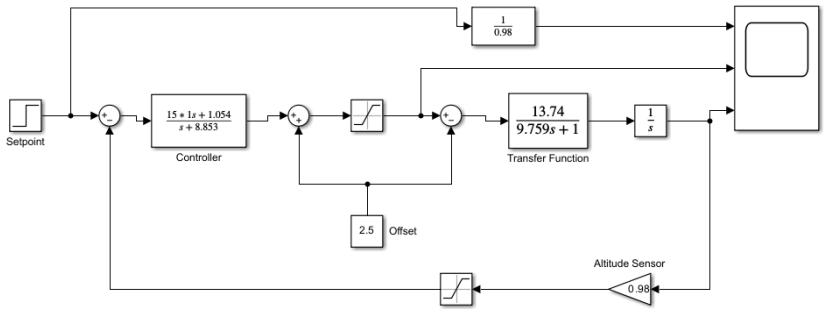

Figure 4: Simulink model showing the lead compensator circuit.

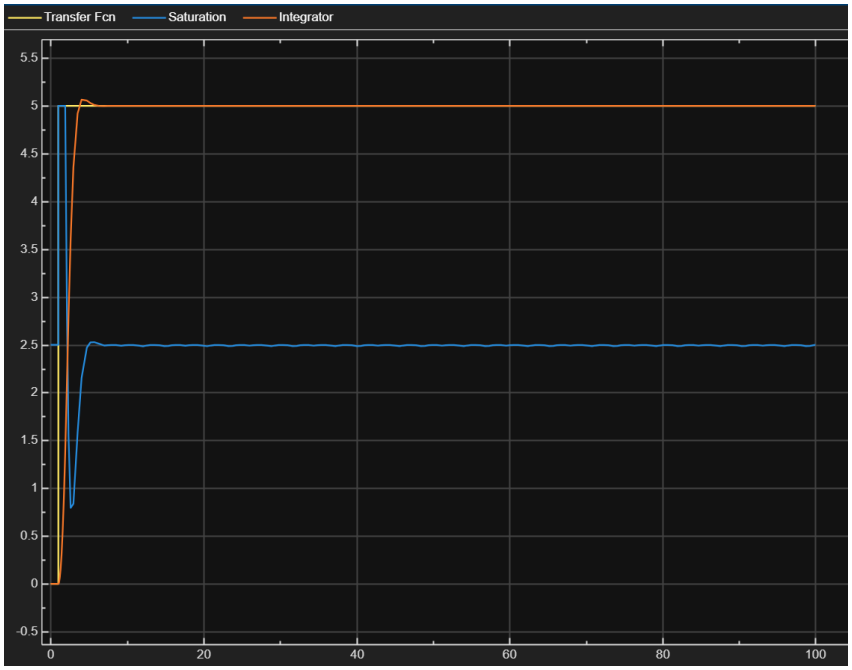

Figure 5: Step response showing steady-state tracking.

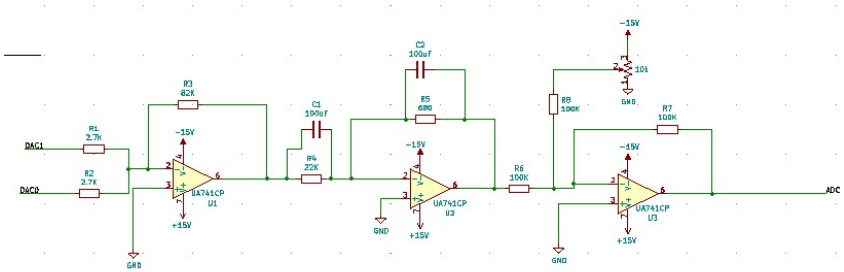

Figure 6: Physical implementation of the lead compensator circuit.

4.3 Settling Time Specification

The required settling time t_e is 8 seconds. The system reaches steady-state and settles within 6.8 seconds. To meet the requirement of settling within 2% of the final value, we calculate 2% of 6.5 seconds:

$$2\% of 6.5 seconds = 0.13 \ seconds$$

Thus, the settling time within 2% is:

$$6.5 + 0.13 = 6.63 \ seconds$$

Since 6.5 seconds is less than the required t_e of 8 seconds, the system successfully meets the settling time specification.

4.4 Disturbance Rejection Specification

A disturbance was applied at around 23 seconds, and the system stabilized and returned to steady-state at around 29.5 seconds, giving a disturbance rejection time of 6. seconds. The requirement is to settle back within 2% of the pre-disturbance value, and using the same calculation for 2%:

$$6.5 + 0.13 = 6.13 \ seconds$$

Since the disturbance rejection time of 6.5 seconds is within this 2% margin (6.13 seconds), and is still less than $t_e = 8$ seconds, the system meets the disturbance rejection requirement.

4.5 Sinusoidal Tracking Specification

When a sinusoidal disturbance with a frequency of 0.1 rad/s was applied at around 73 seconds, the system tracked the sinusoidal input with an error of approximately 2% = 0.1 m. This error is well below the allowed tracking error of 8%, indicating that the system effectively tracks the sinusoidal input within the required tolerance.

Overall, the system performs as expected and satisfies all design specifications, meeting the required settling time and disturbance rejection criteria, as well as sinusoidal tracking performance.

5 Controller Testing

After assembling the lead compensator using the calculated component values from Lab 2, the circuit was initially tested on a breadboard to verify correct functionality. Following multiple iterations of adjustment, the final design was soldered onto a veroboard to provide a stable and reliable platform for further testing.

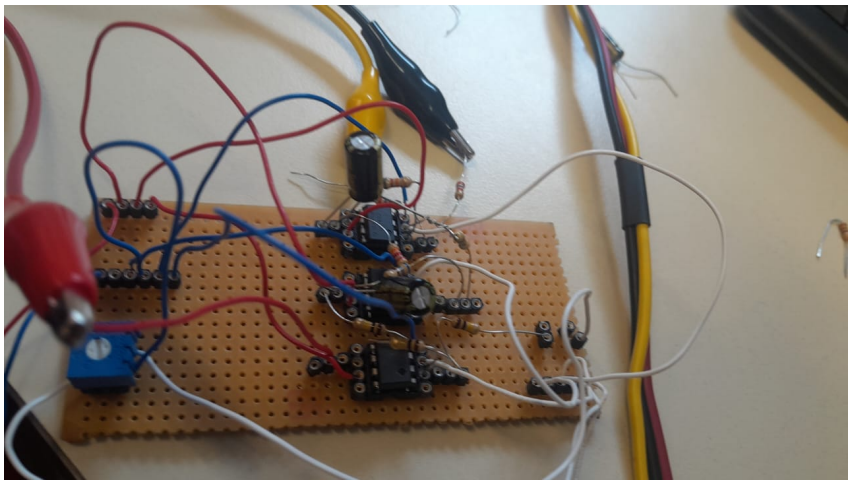

Figure 7: Circuit diagram on a veroboard.

The completed controller was then evaluated using the Control Laboratory Helicopter Project Control 2022 Release 1.4 simulation. The simulation results, shown in Figure 8, illustrate the system's response in terms of tracking accuracy, overshoot, and settling time. These results confirm that the compensator effectively improves the system's dynamic performance and meets the design specifications.

The simulation validates the compensator design, demonstrating improved transient response and ensuring the output follows the desired reference input closely.

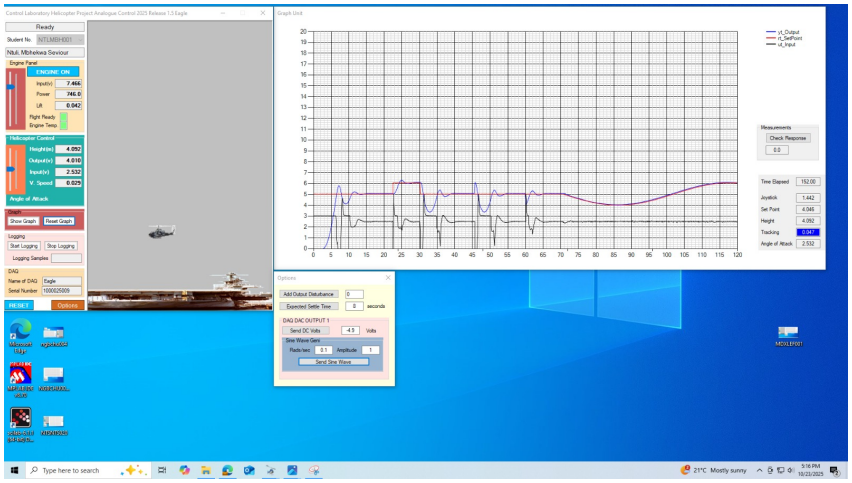

Figure 8: Simulation results demonstrating the kitticopter response with the implemented lead compensator.

6 Conclusion

The implementation of the lead compensator in this experiment successfully improved both the transient and steady-state performance of the kitticopter system compared to the initial proportional controller. The proportional controller established in Lab 2 was not able to achieved an overshoot of only 4%, and shortened the settling time to approximately 6.13 s—all of which satisfy the required system performance specifications but this was found in this expriment does.

The compensator's effectiveness lies in its ability to shift the closed-loop poles further left in the s-plane, thereby increasing the phase margin and damping ratio. This resulted in a system response that is faster, smoother, and more stable. The improved phase margin also contributed to enhanced robustness against disturbances and parameter variations, ensuring the system maintained accurate tracking and quick recovery even under external perturbations.

Furthermore, the lead compensator demonstrated excellent disturbance rejection and sinusoidal tracking performance, maintaining steady operation with minimal phase lag and amplitude error. These results confirm that the compensator not only meets but exceeds the expected design targets. The experiment thus highlights the critical role of compensator design in achieving precise control of dynamic systems. Overall, the study validated the theoretical design process and showcased how control theory principles can be effectively applied to enhance system performance in practice.

7 Recommendations

Although the designed compensator successfully fulfilled all performance requirements, several refinements could be implemented to achieve even better control performance and robustness in future iterations.

7.1 Adjusting Pole–Zero Ratio

The ratio between the compensator pole and zero was determined as:

$$\frac{p}{z} = \frac{8.92}{1.05} \approx 8.49$$

While this ratio was adequate for the current design, slightly reducing the spacing between the pole and zero could further decrease overshoot and improve stability without adding significant complexity. This adjustment could yield a smoother transient response with reduced oscillatory behaviour.

7.2 Cascaded Lead Compensator

For enhanced control precision, a cascaded lead compensator configuration could be explored. Adding an additional lead stage would allow for greater manipulation of phase

margin and transient characteristics. However, this approach should be considered carefully, as it increases the implementation complexity and component sensitivity.

7.3 Implementation of PID Control

Future work could also involve extending the compensator design into a full Proportional–Integral–Derivative (PID) controller. The integral term would help eliminate any residual steady-state error, while the derivative term could improve the response to rapidly changing inputs. A PID configuration would thus provide superior adaptability and accuracy for more demanding applications.

7.4 Gain Optimization and Robustness Enhancement

Although the current gain settings provided stable and accurate performance, further optimization could be conducted to account for component tolerances and real-world disturbances. Fine-tuning the gain in practical conditions would ensure consistent stability and reliability, even under parameter variations or environmental uncertainties.

7.5 Advanced Filtering and State Estimation

To improve disturbance rejection further, advanced estimation techniques such as Kalman filtering could be integrated. These methods would enable faster disturbance detection and suppression, reducing system sensitivity to measurement noise or external disturbances.

7.6 Frequency Response Evaluation

Lastly, the sinusoidal tracking performance was verified at a frequency of 0.1 rad/s. Extending this analysis to higher frequencies would help determine the system's bandwidth and frequency response limitations. A broader bandwidth would enable the compensator to handle more dynamic input variations, making it more suitable for real-world control applications.

In summary, while the lead compensator met all experimental design objectives, the proposed refinements—such as adjusting the pole–zero ratio, optimizing gain, incorporating PID control, and applying advanced filtering—could further enhance the performance, stability, and versatility of the system in future implementations.

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